

Multiple Choice (10 marks)

1. Higher dietary sodium (salt) intake is generally associated with:
 - (a) Decreased calcium excretion in the urine
 - (b) Increased risk of fracture
 - (c) Decreased dietary calcium absorption
 - (d) Negative calcium balance and bone mineral loss
2. The daily amount of protein oxidized in the body of a person spending 24 hours in a respiratory chamber can be determined from his/her:
 - (a) O₂ consumption alone
 - (b) CO₂ production alone
 - (c) Breath collection
 - (d) Total urinary nitrogen excretion alone
3. According to the recommendations for the nutritional management of patients with diabetes, the consumption of saturated fat should be:
 - (a) <10% of total daily energy
 - (b) <12%
 - (c) <15%
 - (d) <16%
4. What is the phenotype of a congenital disorder impairing the secretion of leptin?
 - (a) Normal energy intake, normal body weight and hyperthyroidism
 - (b) Obesity, excess energy intake, normal growth and hypoinsulinaemia
 - (c) Obesity, abnormal growth, hypothyroidism, hyperinsulinaemia
 - (d) Underweight, abnormal growth, hypothyroidism, hyperinsulinaemia
5. What enzyme is responsible for copying methylation marks from the parental to the daughter strand of DNA during replication?
 - (a) DNMT 1
 - (b) DNMT 3 a
 - (c) DNMT 3 b
 - (d) DNMT 3 L

True / False (10 marks)

Answer True or False

6. True or False: Starch mainly contains amylose, which is a polymer of fructose linked by beta 1,4-glycosidic bonds.
7. True or False: C 20 and C 22 polyunsaturated fatty acids serve as important precursors for the synthesis of eicosanoids in the body.
8. True or False: Supplementation with long-chain fatty acids EPA and DHA during pregnancy and infancy can enhance visual performance.
9. True or False: n-3 and n-6 polyunsaturated fatty acids and monounsaturated fatty acids have beneficial effects on plasma LDL cholesterol levels.
10. True or False: The sarcomere is the basic functional unit of skeletal muscle tissue responsible for contraction.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how increased insulin secretion in the fed state influences tissue protein synthesis and analyze the underlying mechanisms involved in this process.
12. Discuss the relationship between dietary supplements, particularly high doses of β -carotene, and the increased risk of lung cancer. What factors contribute to this risk?

End of Paper